

- [Synthesis Messages](#)
- [Synthesis Details](#)
- [Resource](#)
 - [Resource Usage Summary](#)
 - [Resource Utilization Summary](#)
- [Timing](#)
 - [Clock Summary](#)
 - [Max Frequency Summary](#)
 - [Detail Timing Paths Informations](#)

Synthesis Messages

Report Title	GowinSynthesis Report
Design File	D:\code\verilog\uart\gowin\fpga_project\src\baud_rate_gen.v D:\code\verilog\uart\gowin\fpga_project\src\uart_rx.v D:\code\verilog\uart\gowin\fpga_project\src\uart_tx.v D:\code\verilog\uart\gowin\fpga_project\src\top_stm_add_led.v
GowinSynthesis Constraints File	---
Tool Version	V1.9.12.01 (64-bit)
Part Number	GW1NR-LV9QN88PC6/I5
Device	GW1NR-9
Device Version	C
Created Time	Sat Jul 11 15:55:31 2026
Legal Announcement	Copyright (C)2014-2025 Gowin Semiconductor Corporation. ALL rights reserved.

Synthesis Details

Top Level Module	top_add
Synthesis Process	Running parser: CPU time = 0h 0m 0.046s, Elapsed time = 0h 0m 0.209s, Peak memory usage = 153.730MB Running netlist conversion: CPU time = 0h 0m 0s, Elapsed time = 0h 0m 0s, Peak memory usage = 0MB Running device independent optimization: Optimizing Phase 0: CPU time = 0h 0m 0s, Elapsed time = 0h 0m 0.024s, Peak memory usage = 153.730MB Optimizing Phase 1: CPU time = 0h 0m 0s, Elapsed time = 0h 0m 0.008s, Peak memory usage = 153.730MB Optimizing Phase 2: CPU time = 0h 0m 0s, Elapsed time = 0h 0m 0.014s, Peak memory usage = 153.730MB Running inference: Inferring Phase 0: CPU time = 0h 0m 0s, Elapsed time = 0h 0m 0.014s, Peak memory usage = 153.730MB Inferring Phase 1: CPU time = 0h 0m 0s, Elapsed time = 0h 0m 0.008s, Peak memory usage = 153.730MB Inferring Phase 2: CPU time = 0h 0m 0s, Elapsed time = 0h 0m 0.003s, Peak memory usage = 153.730MB Inferring Phase 3: CPU time = 0h 0m 0s, Elapsed time = 0h 0m 0.003s, Peak memory usage = 153.730MB Running technical mapping: Tech-Mapping Phase 0: CPU time = 0h 0m 0s, Elapsed time = 0h 0m 0.011s, Peak memory usage = 153.730MB Tech-Mapping Phase 1: CPU time = 0h 0m 0s, Elapsed time = 0h 0m 0.005s, Peak memory usage = 153.730MB Tech-Mapping Phase 2: CPU time = 0h 0m 0s, Elapsed time = 0h 0m 0.006s, Peak memory usage = 153.730MB Tech-Mapping Phase 3: CPU time = 0h 0m 0.218s, Elapsed time = 0h 0m 0.474s, Peak memory usage = 170.359MB Tech-Mapping Phase 4: CPU time = 0h 0m 0.078s, Elapsed time = 0h 0m 0.146s, Peak memory usage = 170.359MB Generate output files: CPU time = 0h 0m 0s, Elapsed time = 0h 0m 0.077s, Peak memory usage = 170.359MB
Total Time and Memory Usage	CPU time = 0h 0m 0.342s, Elapsed time = 0h 0m 1s, Peak memory usage = 170.359MB

Resource

Resource Usage Summary

- [Synthesis Messages](#)
- [Synthesis Details](#)
- [Resource](#)
 - [Resource Usage Summary](#)
 - [Resource Utilization Summary](#)
- [Timing](#)
 - [Clock Summary](#)
 - [Max Frequency Summary](#)
 - [Detail Timing Paths Informations](#)

Resource	Usage
I/O Port	10
I/O Buf	10
IBUF	3
OBUF	7
Register	67
DFF	2
DFFE	8
DFFS	2
DFFR	23
DFFRE	32
LUT	92
LUT2	18
LUT3	27
LUT4	47
ALU	22
ALU	22
INV	9
INV	9

Resource Utilization Summary

Resource	Usage	Utilization
Logic	123(101 LUT, 22 ALU) / 8640	2%
Register	67 / 6693	2%
--Register as Latch	0 / 6693	0%
--Register as FF	67 / 6693	2%
BSRAM	0 / 26	0%

Timing

Clock Summary:

NO.	Clock Name	Type	Period	Frequency(MHz)	Rise	Fall	Source	Master	Object
1	clk	Base	20.000	50.000	0.000	10.000			clk_ibuf/I

Max Frequency Summary:

NO.	Clock Name	Constraint	Actual Fmax	Logic Level	Entity
1	clk	50.000(MHz)	112.875(MHz)	6	TOP

Detail Timing Paths Information

Path 1

Path Summary:

Slack	11.141
Data Arrival Time	9.185
Data Required Time	20.326
From	num2_0_s2
To	num2_5_s2

- [Synthesis Messages](#)
- [Synthesis Details](#)
- [Resource](#)
 - [Resource Usage Summary](#)
 - [Resource Utilization Summary](#)
- [Timing](#)
 - [Clock Summary](#)
 - [Max Frequency Summary](#)
 - [Detail Timing Paths Informations](#)

Launch Clk	clk[R]
Latch Clk	clk[R]

Data Arrival Path:

AT	DELAY	TYPE	RF	FANOUT	NODE
0.000	0.000				clk
0.000	0.000	tCL	RR	1	clk_ibuf/I
0.000	0.000	tINS	RR	67	clk_ibuf/O
0.726	0.726	tNET	RR	1	num2_0_s2/CLK
1.184	0.458	tC2Q	RF	3	num2_0_s2/Q
2.144	0.960	tNET	FF	2	n129_s/I1
3.189	1.045	tINS	FF	1	n129_s/COUT
3.189	0.000	tNET	FF	2	n128_s/CIN
3.752	0.563	tINS	FF	1	n128_s/SUM
4.712	0.960	tNET	FF	2	n190_s0/I0
5.670	0.958	tINS	FF	1	n190_s0/COUT
5.670	0.000	tNET	FF	2	n189_s/CIN
6.233	0.563	tINS	FF	1	n189_s/SUM
7.193	0.960	tNET	FF	1	n210_s2/I0
8.225	1.032	tINS	FF	1	n210_s2/F
9.185	0.960	tNET	FF	1	num2_5_s2/D

Data Required Path:

AT	DELAY	TYPE	RF	FANOUT	NODE
20.000	0.000				clk
20.000	0.000	tCL	RR	1	clk_ibuf/I
20.000	0.000	tINS	RR	67	clk_ibuf/O
20.726	0.726	tNET	RR	1	num2_5_s2/CLK
20.326	-0.400	tSu		1	num2_5_s2

Path Statistics:

Clock Skew:	0.000
Setup Relationship:	20.000
Logic Level:	6
Arrival Clock Path Delay:	cell: 0.000, 0.000%; route: 0.726, 100.000%
Arrival Data Path Delay:	cell: 4.161, 49.188%; route: 3.840, 45.394%; tC2Q: 0.458, 5.418%
Required Clock Path Delay:	cell: 0.000, 0.000%; route: 0.726, 100.000%

Path 2

Path Summary:

Slack	11.141
Data Arrival Time	9.185
Data Required Time	20.326
From	num1_0_s2
To	num1_5_s2
Launch Clk	clk[R]
Latch Clk	clk[R]

Data Arrival Path:

- [Synthesis Messages](#)
- [Synthesis Details](#)
- [Resource](#)
 - [Resource Usage Summary](#)
 - [Resource Utilization Summary](#)
- [Timing](#)
 - [Clock Summary](#)
 - [Max Frequency Summary](#)
 - [Detail Timing Paths Informations](#)

AT	DELAY	TYPE	RF	FANOUT	NODE
0.000	0.000				clk
0.000	0.000	tCL	RR	1	clk_ibuf/I
0.000	0.000	tINS	RR	67	clk_ibuf/O
0.726	0.726	tNET	RR	1	num1_0_s2/CLK
1.184	0.458	tC2Q	RF	3	num1_0_s2/Q
2.144	0.960	tNET	FF	2	n29_s/I1
3.189	1.045	tINS	FF	1	n29_s/COUT
3.189	0.000	tNET	FF	2	n28_s/CIN
3.752	0.563	tINS	FF	1	n28_s/SUM
4.712	0.960	tNET	FF	2	n90_s0/I0
5.670	0.958	tINS	FF	1	n90_s0/COUT
5.670	0.000	tNET	FF	2	n89_s/CIN
6.233	0.563	tINS	FF	1	n89_s/SUM
7.193	0.960	tNET	FF	1	n111_s2/I0
8.225	1.032	tINS	FF	1	n111_s2/F
9.185	0.960	tNET	FF	1	num1_5_s2/D

Data Required Path:

AT	DELAY	TYPE	RF	FANOUT	NODE
20.000	0.000				clk
20.000	0.000	tCL	RR	1	clk_ibuf/I
20.000	0.000	tINS	RR	67	clk_ibuf/O
20.726	0.726	tNET	RR	1	num1_5_s2/CLK
20.326	-0.400	tSu		1	num1_5_s2

Path Statistics:

Clock Skew:	0.000
Setup Relationship:	20.000
Logic Level:	6
Arrival Clock Path Delay:	cell: 0.000, 0.000%; route: 0.726, 100.000%
Arrival Data Path Delay:	cell: 4.161, 49.188%; route: 3.840, 45.394%; tC2Q: 0.458, 5.418%
Required Clock Path Delay:	cell: 0.000, 0.000%; route: 0.726, 100.000%

Path 3

Path Summary:

Slack	11.679
Data Arrival Time	8.647
Data Required Time	20.326
From	num2_0_s2
To	num2_4_s2
Launch Clk	clk[R]
Latch Clk	clk[R]

Data Arrival Path:

AT	DELAY	TYPE	RF	FANOUT	NODE
0.000	0.000				clk
0.000	0.000	tCL	RR	1	clk_ibuf/I

- [Synthesis Messages](#)
- [Synthesis Details](#)
- [Resource](#)
 - [Resource Usage Summary](#)
 - [Resource Utilization Summary](#)
- [Timing](#)
 - [Clock Summary](#)
 - [Max Frequency Summary](#)
 - [Detail Timing Paths Informations](#)

0.000	0.000	tINS	RR	67	clk_ibuf/O
0.726	0.726	tNET	RR	1	num2_0_s2/CLK
1.184	0.458	tC2Q	RF	3	num2_0_s2/Q
2.144	0.960	tNET	FF	2	n129_s/I1
3.189	1.045	tINS	FF	1	n129_s/COUT
3.189	0.000	tNET	FF	2	n128_s/CIN
3.752	0.563	tINS	FF	1	n128_s/SUM
4.712	0.960	tNET	FF	2	n190_s0/I0
5.695	0.983	tINS	FF	1	n190_s0/SUM
6.655	0.960	tNET	FF	1	n211_s2/I0
7.687	1.032	tINS	FF	1	n211_s2/F
8.647	0.960	tNET	FF	1	num2_4_s2/D

Data Required Path:

AT	DELAY	TYPE	RF	FANOUT	NODE
20.000	0.000				clk
20.000	0.000	tCL	RR	1	clk_ibuf/I
20.000	0.000	tINS	RR	67	clk_ibuf/O
20.726	0.726	tNET	RR	1	num2_4_s2/CLK
20.326	-0.400	tSu		1	num2_4_s2

Path Statistics:

Clock Skew:	0.000
Setup Relationship:	20.000
Logic Level:	5
Arrival Clock Path Delay:	cell: 0.000, 0.000%; route: 0.726, 100.000%
Arrival Data Path Delay:	cell: 3.623, 45.737%; route: 3.840, 48.477%; tC2Q: 0.458, 5.786%
Required Clock Path Delay:	cell: 0.000, 0.000%; route: 0.726, 100.000%

Path 4

Path Summary:

Slack	11.679
Data Arrival Time	8.647
Data Required Time	20.326
From	num1_0_s2
To	num1_4_s2
Launch Clk	clk[R]
Latch Clk	clk[R]

Data Arrival Path:

AT	DELAY	TYPE	RF	FANOUT	NODE
0.000	0.000				clk
0.000	0.000	tCL	RR	1	clk_ibuf/I
0.000	0.000	tINS	RR	67	clk_ibuf/O
0.726	0.726	tNET	RR	1	num1_0_s2/CLK
1.184	0.458	tC2Q	RF	3	num1_0_s2/Q
2.144	0.960	tNET	FF	2	n29_s/I1
3.189	1.045	tINS	FF	1	n29_s/COUT

- [Synthesis Messages](#)
- [Synthesis Details](#)
- [Resource](#)
 - [Resource Usage Summary](#)
 - [Resource Utilization Summary](#)
- [Timing](#)
 - [Clock Summary](#)
 - [Max Frequency Summary](#)
 - [Detail Timing Paths Informations](#)

3.189	0.000	tNET	FF	2	n28_s/CIN
3.752	0.563	tINS	FF	1	n28_s/SUM
4.712	0.960	tNET	FF	2	n90_s0/I0
5.695	0.983	tINS	FF	1	n90_s0/SUM
6.655	0.960	tNET	FF	1	n112_s2/I0
7.687	1.032	tINS	FF	1	n112_s2/F
8.647	0.960	tNET	FF	1	num1_4_s2/D

Data Required Path:

AT	DELAY	TYPE	RF	FANOUT	NODE
20.000	0.000				clk
20.000	0.000	tCL	RR	1	clk_ibuf/I
20.000	0.000	tINS	RR	67	clk_ibuf/O
20.726	0.726	tNET	RR	1	num1_4_s2/CLK
20.326	-0.400	tSu		1	num1_4_s2

Path Statistics:

Clock Skew:	0.000
Setup Relationship:	20.000
Logic Level:	5
Arrival Clock Path Delay:	cell: 0.000, 0.000%; route: 0.726, 100.000%
Arrival Data Path Delay:	cell: 3.623, 45.737%; route: 3.840, 48.477%; tC2Q: 0.458, 5.786%
Required Clock Path Delay:	cell: 0.000, 0.000%; route: 0.726, 100.000%

Path 5

Path Summary:

Slack	12.005
Data Arrival Time	8.321
Data Required Time	20.326
From	rx_inst/sample_cnt_1_s3
To	rx_inst/bitpos_0_s3
Launch Clk	clk[R]
Latch Clk	clk[R]

Data Arrival Path:

AT	DELAY	TYPE	RF	FANOUT	NODE
0.000	0.000				clk
0.000	0.000	tCL	RR	1	clk_ibuf/I
0.000	0.000	tINS	RR	67	clk_ibuf/O
0.726	0.726	tNET	RR	1	rx_inst/sample_cnt_1_s3/CLK
1.184	0.458	tC2Q	RF	5	rx_inst/sample_cnt_1_s3/Q
2.144	0.960	tNET	FF	1	rx_inst/n370_s3/I1
3.243	1.099	tINS	FF	4	rx_inst/n370_s3/F
4.203	0.960	tNET	FF	1	rx_inst/n366_s1/I1
5.302	1.099	tINS	FF	6	rx_inst/n366_s1/F
6.262	0.960	tNET	FF	1	rx_inst/n187_s10/I1
7.361	1.099	tINS	FF	1	rx_inst/n187_s10/F
8.321	0.960	tNET	FF	1	rx_inst/bitpos_0_s3/D

Data Required Path:

AT	DELAY	TYPE	RF	FANOUT	NODE
20.000	0.000				clk
20.000	0.000	tCL	RR	1	clk_ibuf/I
20.000	0.000	tINS	RR	67	clk_ibuf/O
20.726	0.726	tNET	RR	1	rx_inst/bitpos_0_s3/CLK
20.326	-0.400	tSu		1	rx_inst/bitpos_0_s3

Path Statistics:

Clock Skew:	0.000
Setup Relationship:	20.000
Logic Level:	4
Arrival Clock Path Delay:	cell: 0.000, 0.000%; route: 0.726, 100.000%
Arrival Data Path Delay:	cell: 3.297, 43.408%; route: 3.840, 50.558%; tC2Q: 0.458, 6.034%
Required Clock Path Delay:	cell: 0.000, 0.000%; route: 0.726, 100.000%

- [Synthesis Messages](#)
- [Synthesis Details](#)
- [Resource](#)
 - [Resource Usage Summary](#)
 - [Resource Utilization Summary](#)
- [Timing](#)
 - [Clock Summary](#)
 - [Max Frequency Summary](#)
 - [Detail Timing Paths Informations](#)